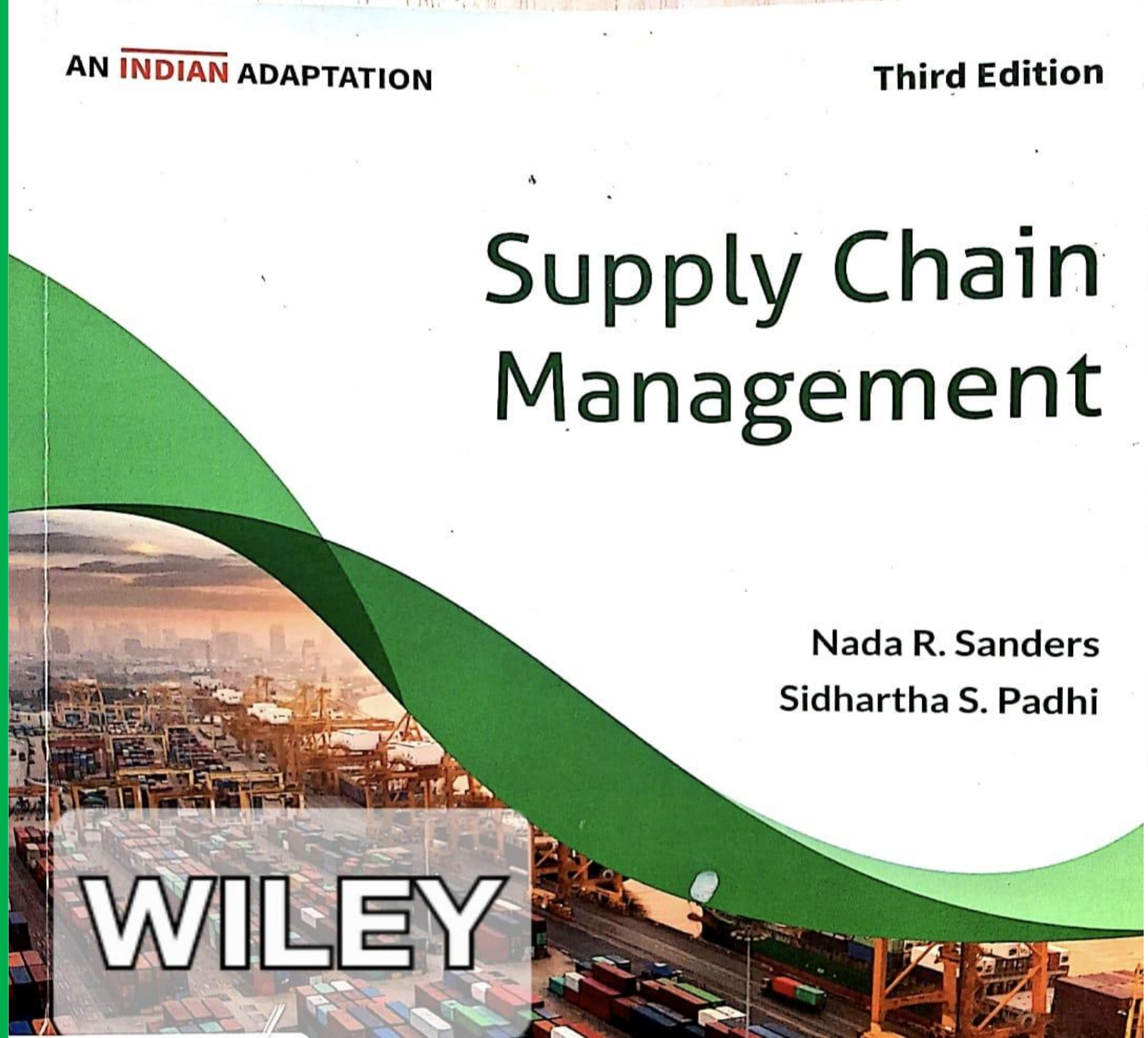
AN INDIAN ADAPTATION
Third Edition
Supply Chain Management
Nada R. Sanders
Sidhartha S. Padhi
WILEY

SCM
SUPPLY CHAIN MANAGEMENT
PROFIT
ANALYSIS
MANAGEMENT
MANUFACTURING
TIME TO MARKET
DISTRIBUTION
PLAN
PROCUREMENT
LOGISTIC

***SUBJECT:* SUPPLY CHAIN MANAGEMENT STRATEGY**

***CHAPTER:* 03**

CHAPTER TITLE: Network and System Design

REFERENCE BOOK:

SUPPLY CHAIN MANAGEMENT (AN INDIAN ADAPTATION), N R SANDERS & S S PADHI, WILEY, 3RD ED. (2023)

Contents

- Processes Across the Supply Chain
- Managing Supply Chain Processes
- Understanding Processes: Theory of Constraints
- Integration of Supply Chain Processes
- Designing Supply Chain Networks
- Enterprise Resource Planning

SECTION 1

Key Concepts

Order Management Process	*It includes managing the customer order from placement to product receipt through the entire chain. This requires process thinking and requires managers to view the collection of process as a system designed to satisfy customer needs.*
Business Process	*It is a set of structured activities or steps with specified outcomes. Consider the process involved in enrolling in a class at your university. There are applications forms to be reviewed by the admission office, financial forms to be complete at the Bursar's office and the class registration to be completed at the Registrar office*
Customer Service Process	*It involves a series of activities designed to enhance the level of customer satisfaction by meeting or exceeding their expectations. This may involve a series of well coordinated activities such as invoicing, handling product returns, providing real-time information on promised shipping dates and product availability.*

Key Concepts

Order Fulfilment Process	*It involves ensuring that customer orders are filled. It may involved activities such as receiving and processing the order, ensuring movement of product and delivery and customer follow-up.*
Manufacturig Process	*It involves ensuring the production of products; the demand management process, which balances demand requirements with operational and supply chain activities.*
Distribution Process	*It involves distributing and delivering products to specified locations.*

Order Management Process

Standard Steps in Order Management Solutions

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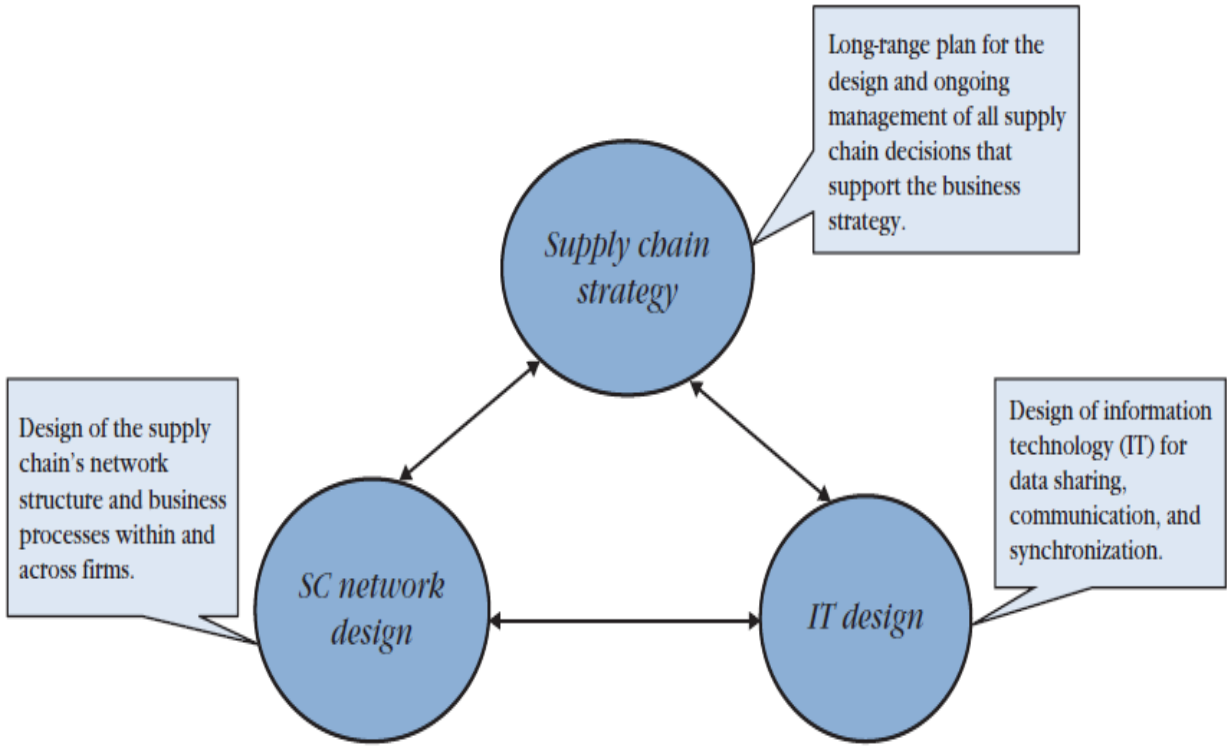
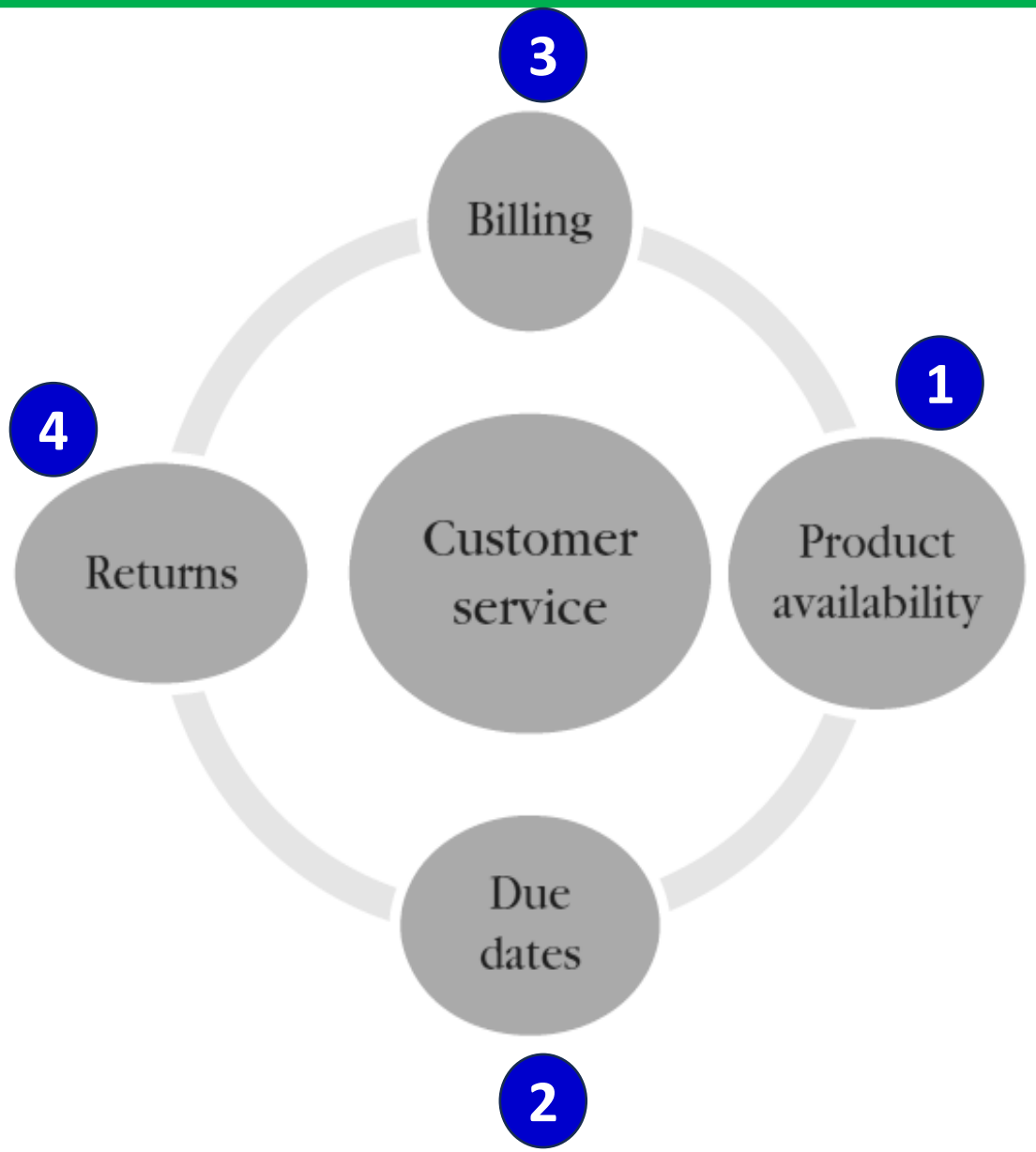
Business Process



- **Supply Chain Strategy**
- This comes first.
- It defines the overall goals: cost leadership, responsiveness, sustainability, customer service levels, etc.
- Strategy sets the direction for how the supply chain should operate to support business objectives.
- **Supply Chain Network Design**
- Once the strategy is clear, the network design follows.
- This involves decisions on facility locations, distribution centers, warehouses, suppliers, and transportation routes.
- The design translates strategic goals into a physical and operational structure.
- **IT Design**
- Finally, IT systems are designed to support and enable the chosen network.
- This includes ERP systems, supply chain management software, data analytics, and integration platforms.
- IT ensures visibility, coordination, and efficiency across the supply chain.

Customer Service Process

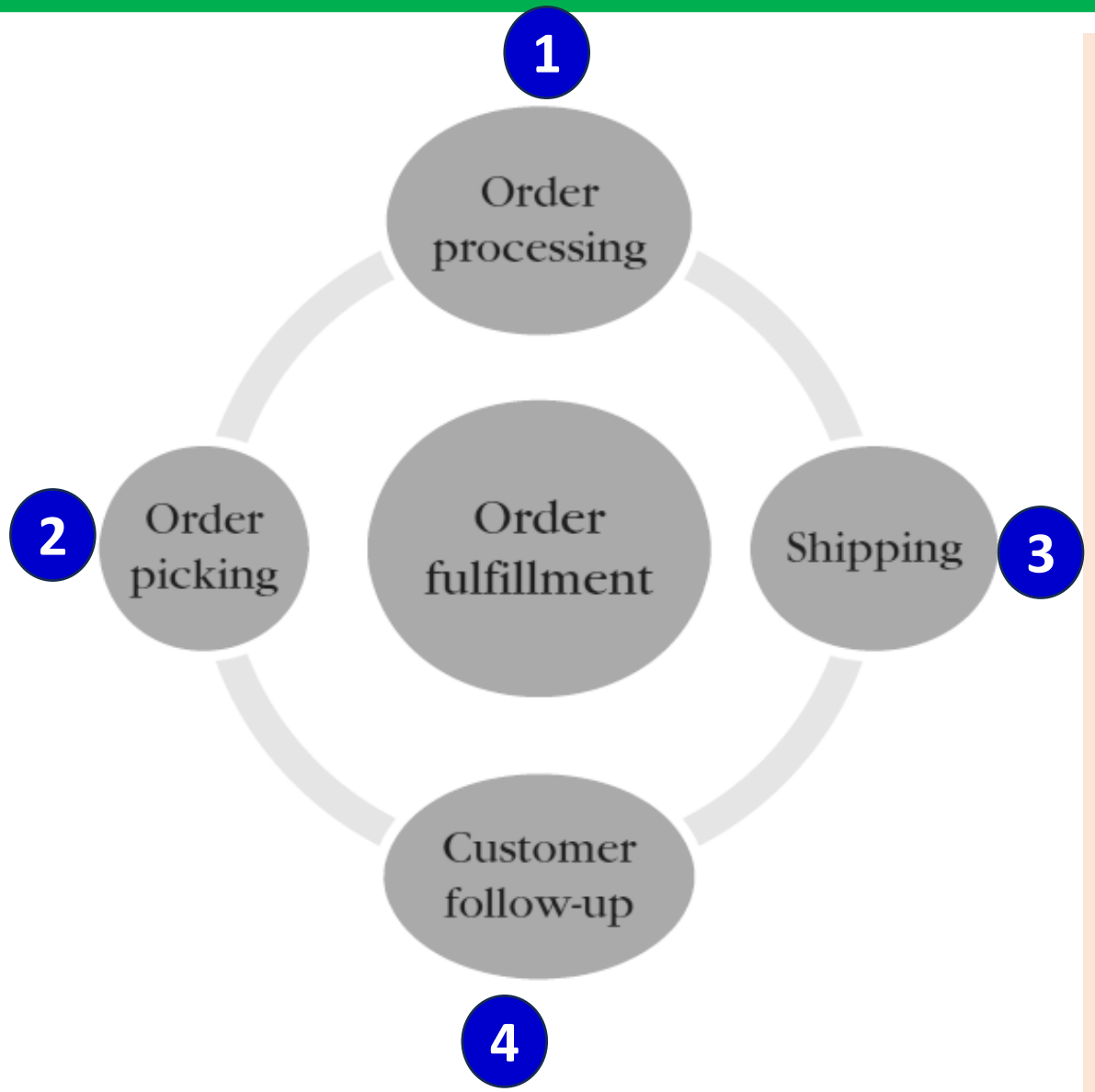


Product Availability

- First, check if the product is in stock or available.
- This avoids unnecessary steps if the item cannot be provided.

- **Due Dates**
  - Next, confirm timelines: delivery dates, payment deadlines, or return windows.
  - Customers need clarity on when they'll receive the product or when actions are required.
- **Billing**
  - Once availability and due dates are clear, proceed with billing.
  - This ensures the customer pays for something that is confirmed and scheduled.
- **Returns**
  - Finally, handle return policies or requests.
  - Returns are usually the last step, triggered only after billing and product delivery

Order Fulfilment Process



Order processing- Once a customer places an order, the order is checked for accuracy, and product availability is confirmed.

Picking Items: A picking team or automated systems retrieve the items according to the packing slip instructions.

Shipping: The final step involves shipping the packed orders to customers. This includes choosing the appropriate shipping method, generating tracking information and ensuring timely delivery.

Customer follow-up: By following up with customers after shipping, businesses can ensure that their customers are informed and satisfied with their order, which can lead to repeat business and positive referrals.

Managing Supply Chain Processes

There are two differing views on how to manage processes across the supply chain, as follows:

Transactional View: It focuses on making supply chain processes more efficient and effective based on quantitative metrics. This can be achieved through supply chain network redesign to promote speed and eliminate redundancy, by standardizing transactions to improve efficiency and by implementing better information technology to improve transfer of information and improve accuracy.

Advantage- The focus is to make supply chain transactions more efficient and effective based on measurable metrics.

Relationship View: It is focused on managing relationship across the supply chain. This involves managing relationships between people and organizations and linking up process across organizations of the supply chain.

Advantage- The objective is to structure inter firm relationships in the supply chain for the longterm benefit of all the supply chain members.

SECTION 2

Understanding Processes:

Theory of Constrains(TOC)

- The TOC process seeks to identify the constraint and restructure to eliminate its impact.
- The system can not produce any more than the output of its bottleneck or constraining activity.
- If we want to increase output, we need to find ways of increasing the capacity of the bottleneck or constraint.
- The premise behind the **Theory of Constraints (TOC)** is that every system has one or more limiting factors that is preventing it from further achieving its goal.
- When the constraint changes, identify the new constraint and continue the steps, as this is a never ending process.

System Constraints

Equipment constraint	Limited by how much it can produce
Labor (people) constraint	Lack of skill or processing capability
Facility constraint	Limited size of the distribution center (warehouse) for storage

The focus of **Theory of Constraints (TOC)** is to uncover the constraint and manage it.

System Constraints – A Study

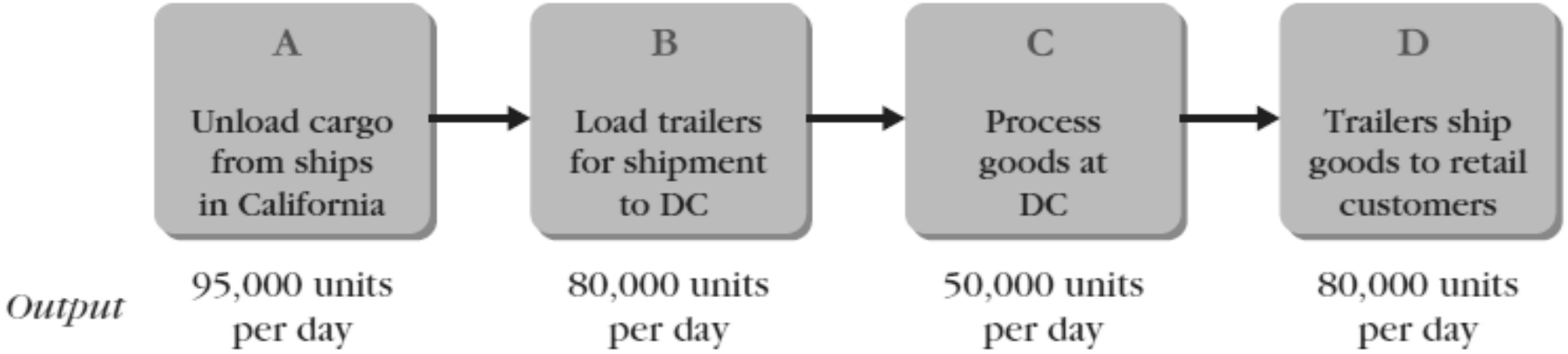
Example: ASCIS' Distribution Network. It had a serial network structure.

Process A (first process), was unloading Cargo from Ships to California ports. Let's assume that this activity had a rate of 95000 units per day.

Process B, load trailers for shipment to the DC, assuming a rate of 80,000 units per day. Then goods are processed at the DC.

Process C, it likely involved the breaking bulk, sorting, and packing at a rate of 50,000 units per day.

Process D (final process), involved shipping trailers of goods to retail customers at a rate of 80,000 units per day.



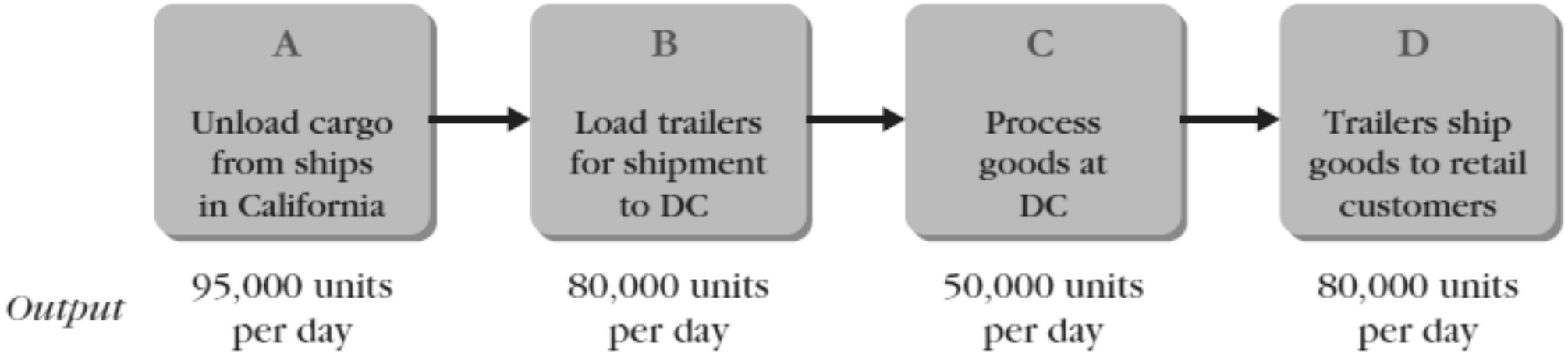
What is the maximum number of units that this system can process in a day?

System Constraints – A Study

What is the maximum number of units that this system can process in a day?

"50,000 units per day or the bottleneck activity"

The reason is that even though processes A and B can handle more units than process C, not more than 50,000 units can go through the bottleneck. This is the constraint.

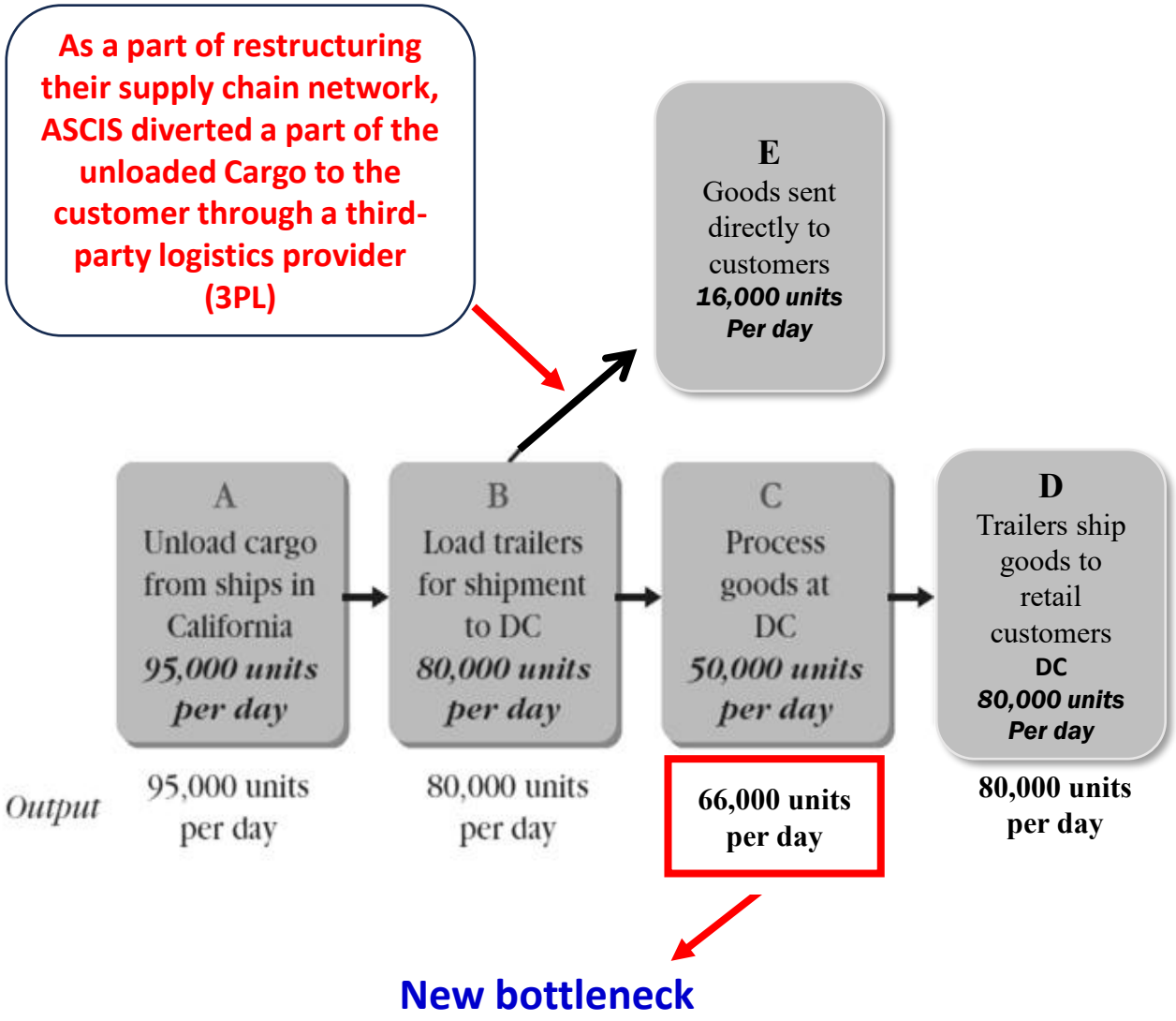


Managing System Constraints

Theory of Constraints (TOC) teaches us to manage the constraint using the following steps:

- **Identify the constraint:** In the ASICS example it was the DC's processing capability.
- **Exploit the Constraint:** Decision must be made on how best to use the constraint and ensure that the constraint's time is not wasted doing things that it should not do.
- **Subordinate all other processes to the above decision:** This means align the whole system around the constraint.
- **Elevate the Constraint:** If possible, increase capacity of the constraint. ASICS bought more capacity by 'bypassing' the DC.
- **When the constraint changes, return to step 1: This means identify the new constraint.**
- **Engage in continuous improvement:** Continue these steps as this is a never-ending process

A Parallel Process:
ASICS Redesigned Distribution Network



- **Stage A (Unload cargo at port)** Capacity: 95,000 units/day → Plenty of capacity, but not the limiting factor.
- **Stage B (Load trailers to DC)** Capacity: 80,000 units/day → Lower than A, so flow reduces here.
- **Stage C (Process goods at DC)** Capacity: 50,000 units/day → This is the **true bottleneck**. Even though the diagram shows an "output" of 66,000 units/day, the processing capacity is only 50,000. That means the system cannot actually sustain 66,000; it will be capped at 50,000.
- **Stage D (Ship goods to retail customers)** Capacity: 80,000 units/day → Higher than C, but since C only passes 50,000, D cannot reach its full potential.
- **Stage E (Direct-to-customer shipments)** Capacity: 16,000 units/day → this is happening in parallel and is enabling a total of 66,000 to pass through the system. **The maximum output of the system is the new bottleneck**, which is 66,000 units per day. This is a **new constraint**.

System Variation

Every system or process has variations. Like Sometimes workers in a factory are very productive and other times, no matters how hard you try, no fruitful service comes from workers. The same variation is true for warehouse, restaurant, office environment or medical facility, equipment and facilities. Sometimes processes run smoothly, whereas other times there are breakdowns requiring unexpected repairs.

Problems of Variation: Variation is a problem as it consumes resources.

Time- Sometimes it takes more time to do an activity.

Space- Needing extra storage.

Labour- Tasks having to be repeated.

Demand Gap- Sometimes there are unexpected surges in demand and other times there are slumps. These variation in demand create variations in production schedules of manufacturers and suppliers, which creates a coordination problem.

Management of Variation

Process variation needs to be managed as it can create significant problems for a system. There are three ways this can be accomplished.

First, attempts should be made to reduce or eliminate as much as variation possible. This requires identification of source of variability in the system such as workers or equipment, and correcting the problem.

Second, create buffers to deal with the variation. Buffers in the form of excess inventories, can be placed before and after highly variable activities. They can be placed before the constraint, ensuring that the constraint is 'starved.' They can also be placed behind the constraint to prevent downstream failure to block the constraints output.

Third, managers can deal with variation by designing more flexibility into the process to respond to the variation. This maybe in the form of flexible technology or cross-training workers.

Capacity Implication

Capacity refers to the maximum amount of output that can be achieved by a process over a specified period of time. Different businesses measure capacity in different ways, all trying to assess a measure of processing capability.

For example, a health care clinic measures capacity in terms of patients seen per day, restaurants in terms of customer served, a delivery company in terms of number of packages delivered and an insurance company in terms of number of claims processed. Capacity can also be measured in terms of size or storage limit. For example, a hospital measures its capacity in terms of numbers of hospital beds available, a restaurant in terms of number of seats available and a warehouse in terms of its square footage.

Capacity Utilisation Formula

$$\text{Utilization} = \left[\frac{\text{Actual Output Rate}}{\text{Capacity}}\right](100\%)$$

Techniques of Measurement of Capacity

Design Capacity- **The maximum output rate that can be achieved by a facility**. A company achieves this output rate by using temporary measures, such as overtime, overstaffing, using equipment at the maximum rate and subcontracting.

Effective Capacity- **The maximum output rate that can be sustained under normal conditions**. These conditions include realistic work schedule and breaks, regular staff levels, schedule machine maintenance, and none of the temporary measures that are used to achieve design capacity.

Capacity Utilization Metrics

(How much of capacity is actually being used)

Example 1: **Computing Capacity Utilization**

Assuming that an automotive manufacturing facility can produce 1000 bikes in a single day, under maximum conditions, using overtime labor. However, the facility was producing 900 bikes per day under normal conditions. During the month of June the facility was producing 800 bikes per day. What is the facility's capacity utilization for both design and effective capacity.

Solution:

$$\text{Utilization}_{\text{effective}} = \left[\frac{\text{Actual Output}}{\text{Effective Capacity}}\right](100\%) = \left[\frac{800}{900}\right](100\%) = 88.9\%$$

$$\text{Utilization}_{\text{design}} = \left[\frac{\text{Actual Output}}{\text{Design Capacity}}\right](100\%) = \left[\frac{800}{1000}\right](100\%) = 80\%$$

The effective utilization rates show that the current output of the manufacturing facility is higher than design utilization. Thus, the facility is utilizing its resources very effectively

Questions (To be solved)

Shex Hotel can accommodate 300 guests in a single day under maximum conditions. However, under normal conditions this hotel is designed to accommodate 200 guests per day. In September, the hotel accommodated 150 guests in one day. What is the hotel's capacity utilization?

Compute:

Utilization $_{effective}$???

Utilization $_{design}$???

Questions (To be solved)

Papi Toys can produce 500 toys in a single day under maximum conditions. However, it is designed to produce 400 units per day under normal conditions. In February, it produced 480 units per day under normal conditions. What was Papi Toys' capacity utilization for both design and effective capacity? What do these numbers tell you?

Compute:

Utilization $_{effective}$???

Utilization $_{design}$???

SECTION 3

Integration of Supply Chain Processes

Stages of Integration: There are three stages of Integration.

Stage 1: Represents completely functional Independence. It means the functions in the company operate in complete isolation of one another, called the “silo” mentality.

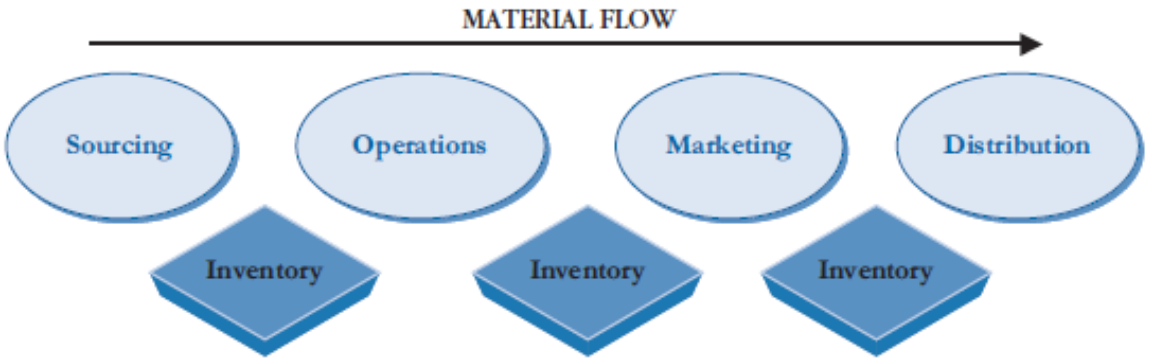
Stage 2: Functions are integrated within the organization. It focuses on internal functional cooperation and coordination.

Stage 3: Functions are integrated across the supply chain. Here the concept of linkages and coordination extends beyond the boundaries of the firm to supply chain partners.

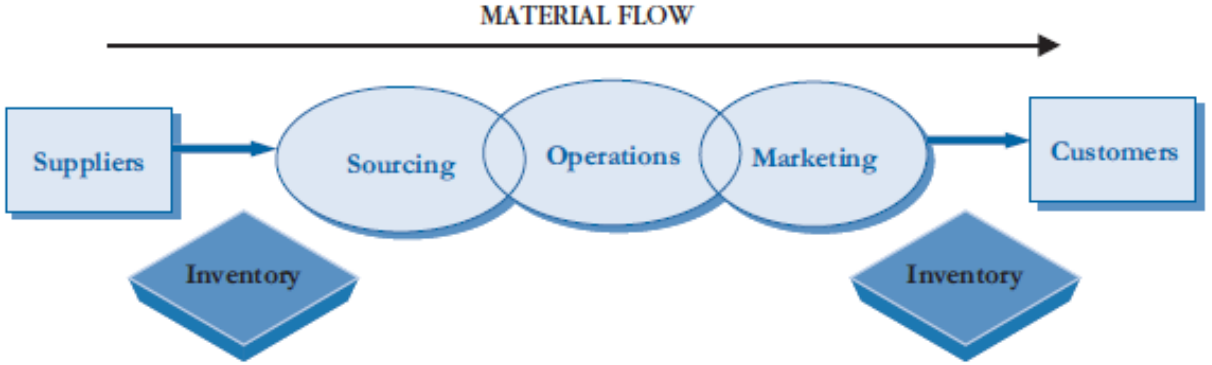
However, companies do not go through this evolution willingly as it requires significant organizational change and restructuring. Rather they typically evolve in response to competitive pressure and threats.

Stages of Supply Chain Integration

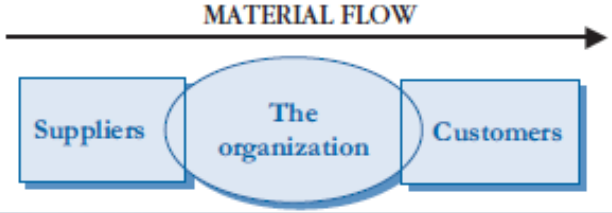
1. Organizational functions are independent.



2. Functions are integrated within the organization.



3. Functions are integrated across the supply chain.



Designing Supply Chain Network

All firms belong to one or more supply chains that begin with the source of raw materials and end with consumption.

There are two aspects of the supply chain network, such as

- ❖First is the actual physical structure of the network itself.
- ❖Second is the management of the network.

Supply Chain Structure and Management

There are countless network structure types and the structure itself evolves based on many factors like types of product (Perishable products, innovative products etc.), available suppliers, and availability of raw materials as well as ease of access to customers.

It is defined by three elements:

1. **The number of companies that are part of the supply chain** -some supply chain are complex and have many members; others have few.
2. **The structural dimensions of the network** – This is the number of tiers in supply chain and how many members are in each tier. A supply chain can have many tiers but few members in each tier, or it can have few tiers but many members in each.
3. **The number of process links across the supply chain**- The more process links that must be managed across the supply chain, the more complex the management of the chain. Example: Customer Relationship Management, Demand Management, order fulfilment or manufacturing flow management.

Designing Segmented Structure

To design segmented structure of supply chain a three step approach can be used as follows

1. **Identify key drivers of operational complexity:** This means identifying what matters most to a particular customer group or channel.

For example-Volume flexibility and supply chain responsiveness are much more important to outlets that attract customer traffic with large , temporary price cuts than they are to " everyday low price" outlets.

Similarly, quick time to market matters much more to fashion oriented, short-life-cycle merchandise than to long lasting, core products.

2. **Design differentiated supply chain segments** tailored to address these unique complexities: Once key drivers of each customer group are identified a supply chain segment must be created to operationalize each one of those characteristics. This will likely require designing multiple supply chain segments to meet the identified requirements. The number of segements, however, should not be too large and should remain reasonable to manage such as five or six.

3. **Create a customized end-to-end operational blue print and performance metrics** for each supply chain segment: Once the segments have been created, details of putting them into operation must be established, including aligning processes and metrics across the supply chain.

Example

- High volume, low-variability demand will likely require a cost-efficient supply chain structure , with dedicated production facilities to drive the lowest cost.
- High volume, high-variability demand, on the other hand, is less predictable.
- High priority customers may be offered a responsive service, where as less profitable segments are offered more limited options, helping to reduce variability in that segment.
- Low volume, low-variability products may be shipped to high priority customers using responsive inventory system, while limiting choices for lower priority customers.
- By segmenting supply chain , a company can offer increased responsiveness for fast growing channels and product lines while providing better cost control in mature and stable segments.

SECTION 4

IT as an enabler of SCM

- Information Technology (IT) enables communication, storage and processing of data within and between firms.
- It is also used to organise information to help managers with decision making.
- To manage its inventory, Wal-Mart uses an IT system called Retail Link, which enables suppliers to see the exact number of their products on every shelf of every store.
- Enterprise resource planning has enabled sharing of information and data across the enterprise for better decision making.

ERP: Enterprise Resource Planning

ERP systems are fully integrated computer based technology used by organisations to manage resources throughout the supply chain.

Purpose: To facilitate the flow of information between all business functions inside the boundaries of the organisation and manage the connections

ERP system typically tie together all business processes, such as sales, production, inventory management, warehousing, delivery, accounting, billing and human resource management.

ERP systems are cross functional and enterprise wide, connecting functions such as marketing, operations, sourcing, and logistics.

ERP: Configuration

ERP configuration is all about **customizing** your **Enterprise Resource Planning Software** to meet your business's unique **operational requirements**. You adjust settings like time zones, currencies, and workflows to align with your processes, enhancing efficiency and productivity across departments.

Unlike customization, configuration is quicker, cost-effective, and typically preserves its settings during software updates.

It allows your internal IT team to make changes without altering the source code, supporting **compliance and integration** with other software tools

Different companies select different configuration based on their needs and resources. The software is developed in components or modules, specific to each function or process.

Components of these modules are shown below fig 3.18

Enterprise Resource Planning (ERP) Modules

Manufacturing

- Bills of material
- Scheduling
- Capacity
- Workflow
- Quality control
- Cost management
- Manufacturing process

Supply Chain Management

- Inventory
- Order entry
- Purchasing
- Supply chain planning
- Supplier scheduling
- Product inspection

Finance

- General ledger
- Cash management
- Accounts payable
- Accounts receivable
- Fixed assets

Project Management

- Costing
- Billing
- Time expense
- Activity management

Human Resources (HR)

- Payroll
- Training
- Hourly involvement
- Employee tracking
- Benefits

Customer Relationship Management (CRM)

- Sales and Marketing
- Commissions
- Service
- Customer contact
- Customer information
- Call Center support

ERP Implementation

ERP implementation is the process of setting up and configuring an ERP system to align with a business's specific needs. It involves various steps, including software installation, data migration, user training, and system testing. Proper implementation ensures that the ERP system functions effectively and meets the company's requirements. An ERP system adds to your competitive advantage by synchronizing all of your information and streamlining workflows.

ERP implementation has proven benefits such as-

- Enhanced business reporting
- Better customer service
- Improved inventory costs
- Better cash flow
- Cost savings
- Better data security
- Business process improvements
- Supply chain management

7 Steps to a Successful ERP Implementation

1

Identification: Define your goals and requirements, know exactly what objectives it can assist you in achieving

2

Selection: Select a vendor who is an ERP expert with in-depth experience, and a project team with skill sets and excellent communication skills

3

Design: Emphasize developing a system configuration that maximizes the client's system benefits and ROI

4

Development and testing: The IT team and ERP vendor implementation team get the software ready and test it with dummy data

5

Training: Focus on comprehensive training for the trainers, allow users to provide feedback and the implementation team to take action on it

6

Deployment: Choose from a big bang adoption, phased adoption, or a parallel adoption for the final deployment

7

Ongoing support: Invest time and resources to detect problems and correct errors throughout the ERP solution's life cycle

Example: ERP Implementation

- Starbucks Corporation, a renowned chain of coffeehouses and roastery reserves, utilizes Oracle ERP, a cloud-based software system that streamlines back-office tasks and daily business operations.
- With the support of Oracle E-Business Suite, Starbucks has access to integrated Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), and Supply Chain Management (SCM) functionalities.
- Oracle ERP enables Starbucks to analyze revenue data, view expenses and sales information, all on a single screen. This centralized system provides convenient and quick access to crucial business information.

Thank You